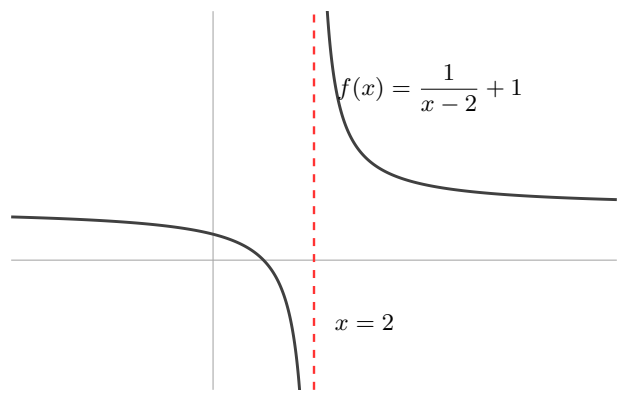




A graph of the function $f(x) = \frac{1}{x-2} + 1$ is shown on a Cartesian coordinate system. The x-axis and y-axis are represented by thin grey lines. A vertical dashed red line at $x = 2$ indicates a vertical asymptote. The function is plotted as a solid black curve that approaches the asymptote from both sides. For $x < 2$, the curve is in the lower-left region, passing through the x-axis at $x = 3$ and approaching negative infinity as x approaches 2 from the left. For $x > 2$, the curve is in the upper-right region, approaching positive infinity as x approaches 2 from the right and approaching the horizontal asymptote $y = 1$ as x increases. The equation $f(x) = \frac{1}{x-2} + 1$ is written in black text in the upper right quadrant, and the equation $x = 2$ is written in black text near the bottom of the dashed red line.

$$f(x) = \frac{1}{x-2} + 1$$

$$x = 2$$